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MIT SCHOOL OF ENGINEERING

Rajbaug, Loni-Kalbhor, Pune


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Subject: UCD

Class: TY-CORE 6

Assignment-1Q1.
Soln.

a) Distinguish b/w Good and Poor Design.

Aspect	Good Design	Bad/Poor Design
• Usability	Easy to use, intuitive, require minimal learning.	Difficult to use, confusing, steep learning curve.
• Functionality	Fulfills its intended purpose effectively and efficiently.	Ineffective, may have unnecessary features or lack essential ones.
• User-Centered	Focuses on user needs, preferences and limitations	Ignores user needs, not tailored to the target audience.
• Efficiency	Enables user to perform tasks quickly and easily	Inconsistent Time-consuming, require unnecessary steps.

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|-------------|--|--|
| Consistency | Uniform design elements, predictable behaviour | Inconsistent, unpredictable, lacks uniformity. |
|-------------|--|--|

Q1. b) Explain what is Interaction Design and give role of it in HCI.

- Interaction Design (IXD) is the process of designing interactive digital products and system with a focus on how users interact with them. It aims to create interface that are intuitive, efficient, and user-friendly, ensuring users can achieve their goal with ease.

Role of Interaction in HCI

- Enhancing Usability: Making system easy and intuitive to use.
- Providing Feedback: Ensuring users understand the system's responses to their actions.
- Ensuring Consistency: Creating a uniform experience across different platforms and contexts.
- Improving Accessibility: Designing interfaces that are accessible to all users, including those with disabilities.
- Increasing User Satisfaction: Focusing on user need to create enjoyable and effective interactions.



Q2. a) Discuss core characteristics of Interaction Design.

→ Core Characteristics of Interaction Design (IXD)

- 1.) User-Centered! - Focuses on user need and expectations to create intuitive interface.
- 2.) Goal-Oriented! - Design interactions to help users achieve their goal efficiently.
- 3.) Iterative Process! - Continuously refines design based on user feedback & testing.
- 4.) Context-Aware! - Adapts to the environment and devices where the interaction occurs.
- 5.) Feedback-Oriented! - Provides clear, immediate feedback to user's actions.
- 6.) Scalable & Flexible! - Works across different devices and allows customization.
- 7.) Aesthetic & Functional Balance! - Combines visual appeal and function with practicality.
- 8.) Consistency! - Ensures uniform design elements and interaction patterns.

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Q2. b) Explain elements of User Experience (UX)

→ Elements of User-Experience (UX):-

1.) Usability:-

- Refers to how easy and intuitive a product is to use.
- Ensures that users can complete task efficiently with confusion or frustration.

2.) Desirability:-

- The emotional appeal of a product, including its visual design, branding and overall aesthetics.

3.) Accessibility:-

- Ensures the product is usable by everyone, including ~~with~~ those with disabilities.

4.) Utility:-

- The functionality and usefulness of the product.

5.) Findability:-

- How easily user can navigate and find what they need.

6.) Credibility:- The trustworthiness and reliability of the product.

7.)

Value:

- The overall benefit and satisfaction the user gets from the product.

Q3.) a) Discuss Design Principles in details:

- i) Consistency: Ensure uniformity across design elements for a predictable and learnable interface.
- ii) Visibility: Make important functions and information easily accessible and noticeable.
- iii) Feedback: Provides clear, immediate response to user actions to guide their interactions.
- iv) Affordance: Design elements should suggest their purpose (e.g. buttons should look clickable).
- v) Simplicity: Keep the design straightforward to focus on essential tasks and reduce complexity.
- vi) User-Control: Allows users to manage their actions with options to undo, redo & adjust.
- vii) Flexibility: Support customization and accommodate diverse user need and preferences.

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viii) Error Prevention: Minimize user errors with clear instructions and easily/easy recovery options.

Q3. b) Discuss importance of Colors in IxD.
→ Importance of Colors in IxD:

i) Visual Hierarchy: Guides user attention and organizes information.

ii) Brand Identity: Reinforces branding and creates a cohesive look.

iii) Emotional Impact: Influences user emotions and enhances the experiences.

iv) Feedback & Affordance: Indicate states and signal interactivity.

v) Accessibility: Ensures readability and inclusivity for all users.

vi) Cultural Significance: Respects cultural difference and avoids misinterpretation.

vii) Consistency: Maintains design harmony & aids ~~consis~~ usability.